

Question number	Answer	Marks
6(a)	An explanation that includes the following points: • many small alveoli to provide a large surface area to increase the rate of diffusion (1) • thin epithelium to increase rate of diffusion (1) • good blood supply to maintain diffusion gradient (1)	(3)
Question number	Answer	Mark
6(b)(i)	• a version of a gene	(1)

Question number	Answer	Mark
6(b)(ii)	An explanation that includes the following points: • there will be a different sequence of R groups (1) • therefore the CFTR protein has a different tertiary structure (1) • because of different {types of / position of} bonds between the R groups (1) • therefore the movement of chloride ions through the cell membrane is affected (1)	(4)

Question number	Answer	Mark
6(b)(iii)	An explanation that includes any four of the following points: • produces very thick, sticky mucus (1) • because of reduced water transport from cells (1) • cilia lining airways are unable to move mucus (1) • therefore microorganisms get trapped in the mucus (1) • mucus provides suitable growth conditions for growth of microorganisms (1)	(4)